Many-body perturbation theory: diagram rules, third order and the linked-diagram theorem

FYS4480/FYS9480 – Week 45, November 5–6, 2026

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Thursday: resolvents, Wick contractions, symmetry factors and fermion signs
Friday: the fourteen third-order diagrams, Møller–Plesset third order, and the pairing model

Reading: lecture notes chapter 9, section 9.15 (diagrammatic approach and the linked diagram theorem); Shavitt and Bartlett, chapters 2–5. Code: `BookPrograms/chapter09/diagrams.py`; notebook `week45.ipynb`.

Answers to the week-44 exercises and the week-45 exercises: last frames.

Plan for week 45

Thursday, 2×45 min: the lists, and the rules derived

- From week 44: the recursion, $\Omega^{(1,2,3)}$ and $E^{(1,2,3,4)}$ in resolvent form and in the determinant basis
- The scalar separation $H_I = E^{(1)} I + W$ absorbs the explicit $E^{(1)}$ insertions in the recursion. $E^{(3)} = \langle W R_0 W R_0 W \rangle_0$
- Contractions are lines; cuts are denominators ($[H_0, a_p^\dagger] = \varepsilon_p a_p^\dagger$); what Q removes
- The weight (4^{nv} relabellings, equivalent lines) and the sign (vertex ends, the permutation π, $(-1)^{K-I}(-1)^p = (-1)^{h+l}$)
- First and second order re-read with the rules

Friday, 2×45 min. Third order, enumerated and evaluated

- Lecture 3 enumerates fourteen connected diagrams. Three give MP3 (pp ladder, hh ladder, ph ring). Eleven carry one-body vertices.
- Lecture 4 evaluates the diagrams for the pairing model. $-53/576 - 53/576 + 29/288 = -1/12$ in units of g^3, and where the "renormalisation term" went; reading a diagram into code
- A look ahead to unlinked fourth-order diagrams. Week 46 develops the linked-diagram theorem and coupled cluster.

Where we are, and the route through this week

Done

- Full CI (weeks 37–38): exact, exponential; the amplitude equations drawn as diagrams with four rules we did not prove
- Hartree-Fock (weeks 38–39): $f_{ai} = 0$
- Week 44: P, Q, the resolvent, BW versus RS, $E^{(2)}$ from Wick's theorem, a two-level benchmark

This week

1. The complete lists through fourth order
2. The four rules, with proofs
3. Third order: fourteen diagrams
4. Third order evaluated in the pairing model

Many-body connection

Notation as in week 44: $H = H_0 + H_I$ with $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$, reference $|\Phi_0\rangle$ with $\varepsilon_0 = \sum_i \varepsilon_i$, $R_0 = Q/(\varepsilon_0 - H_0)$, $D_m = \varepsilon_0 - \varepsilon_m$, $\langle X\rangle_0 = \langle\Phi_0|X|\Phi_0\rangle$, $V_{mn} = \langle\Phi_m|H_I|\Phi_n\rangle$. Occupied i, j, k, l, empty a, b, c, d. Antisymmetrised elements $\overline{v}_{pqrs} = \langle pq|v|rs\rangle_{\mathrm{AS}}$ throughout.

Thursday, lecture 1: The lists, and from sums to pictures

The recursion that generates every order (week 44, lecture 4)

Introduce $H(\lambda) = H_0 + \lambda H_I$, $|\Psi_0(\lambda)\rangle = \sum_{n\geq 0} \lambda^n |\Psi^{(n)}\rangle$ with $|\Psi^{(0)}\rangle = |\Phi_0\rangle$, and $\Delta E(\lambda) = \sum_{n\geq 1} \lambda^n E^{(n)}$. With $R_0 = Q/(\varepsilon_0 - H_0)$, collect powers in

$$|\Psi_0(\lambda)\rangle = |\Phi_0\rangle + R_0[\lambda H_I - \Delta E(\lambda)]|\Psi_0(\lambda)\rangle :$$

$$\boxed{|\Psi^{(n)}\rangle = R_0\Big[H_I|\Psi^{(n-1)}\rangle - \sum_{k=1}^{n-1} E^{(k)}|\Psi^{(n-k)}\rangle\Big], \qquad E^{(n+1)} = \langle\Phi_0|H_I|\Psi^{(n)}\rangle.}$$

- The subtracted terms are the normalisation ("renormalisation", "unlinked") corrections; they appear from $n = 2$ on.
- $R_0|\Phi_0\rangle = 0$: every $|\Psi^{(n)}\rangle$, $n \geq 1$, lies in Q space, so intermediate normalisation is preserved order by order.
- `mbpt.py` implements exactly this recursion and checks it against the explicit expressions on the next two frames.

All contributions to the wave operator through third order

Write $|\Psi_0\rangle = \Omega|\Phi_0\rangle$ with $\Omega P = P + \Omega^{(1)}P + \Omega^{(2)}P + \Omega^{(3)}P + \cdots$, $R \equiv R_0$, $V \equiv H_I$:

$$\Omega^{(1)} = RV,$$

$$\Omega^{(2)} = RVRV \; - \; E^{(1)}R^2V,$$

$$\Omega^{(3)} = RVRVRV \; - \; E^{(1)}\big[RVR^2V + R^2VRV\big] \; + \; \big(E^{(1)}\big)^2R^3V \; - \; E^{(2)}R^2V.$$

In the determinant basis, with $V_{mn} = \langle\Phi_m|H_I|\Phi_n\rangle$ and $D_m = \varepsilon_0 - \varepsilon_m$ $(m, n, I \neq 0)$:

$$\langle\Phi_m|\Psi^{(1)}\rangle = \frac{V_{m0}}{D_m}, \qquad \langle\Phi_m|\Psi^{(2)}\rangle = \sum_n \frac{V_{mn}V_{n0}}{D_mD_n} - \frac{V_{00}V_{m0}}{D_m^2},$$

$$\langle\Phi_m|\Psi^{(3)}\rangle = \sum_{nI} \frac{V_{mn}V_{nI}V_{I0}}{D_mD_nD_I} - V_{00}\Big[\sum_n \frac{V_{mn}V_{n0}}{D_mD_n^2} + \sum_n \frac{V_{mn}V_{n0}}{D_m^2D_n}\Big] + V_{00}^2\frac{V_{m0}}{D_m^3} - E^{(2)}\frac{V_{m0}}{D_m^2}.$$

The displayed operator expressions act on P. For a two-body H_I, $\Psi^{(1)}$ has singles $(\tilde{f}_{ai}/(\varepsilon_i - \varepsilon_a))$ and doubles $(\overline{v}_{abij}/(\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b))$; $\Psi^{(2)}$ reaches up to quadruples; $\Psi^{(3)}$ up to sextuples. Rank is not order.

All contributions to the energy through fourth order

With $\langle X\rangle_0 = \langle\Phi_0|X|\Phi_0\rangle$, $R = Q/(\varepsilon_0 - H_0)$, $V = H_I$, and $E^{(n+1)} = \langle V\Omega^{(n)}\rangle_0$:

$$E^{(1)} = \langle V\rangle_0,$$

$$E^{(2)} = \langle VRV\rangle_0,$$

$$E^{(3)} = \langle VRVRV\rangle_0 \;-\; E^{(1)}\langle VR^2V\rangle_0,$$

$$E^{(4)} = \langle VRVRVRV\rangle_0 \;-\; E^{(1)}\big[\langle VR^2VRV\rangle_0 + \langle VRVR^2V\rangle_0\big] \;+\; \big(E^{(1)}\big)^2\langle VR^3V\rangle_0 \;-\; E^{(2)}\langle VR^2V\rangle_0.$$

In the determinant basis:

$$E^{(1)} = V_{00}, \quad E^{(2)} = \sum_m \frac{V_{0m}V_{m0}}{D_m}, \quad E^{(3)} = \sum_{mn} \frac{V_{0m}V_{mn}V_{n0}}{D_mD_n} - V_{00}\sum_m \frac{V_{0m}V_{m0}}{D_m^2},$$

$$E^{(4)} = \sum_{mnl} \frac{V_{0m}V_{mn}V_{nl}V_{l0}}{D_mD_nD_l} - 2V_{00}\,\mathrm{Re}\sum_{mn} \frac{V_{0m}V_{mn}V_{n0}}{D_m^2D_n} + V_{00}^2\sum_m \frac{V_{0m}V_{m0}}{D_m^3} - E^{(2)}\sum_m \frac{V_{0m}V_{m0}}{D_m^2}.$$

All determinant indices exclude the reference. The two mixed terms are complex conjugates, giving $2\,\mathrm{Re}$. The real part can be omitted for real matrix elements. Terms containing $E^{(k)}$ are products of lower-order quantities. Week 46 develops their cancellation against the unlinked Wick diagrams.

The one-body residual and the second-order energy

Let $H = \sum_{pq} h_{pq} a_p^\dagger a_q + V_2$ and $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$. Define

$$u_{pq} = h_{pq} - \varepsilon_p \delta_{pq}, \quad U_{pq} = \sum_j \overline{v}_{pjqj}, \quad \tilde{f}_{pq} = u_{pq} + U_{pq}.$$

Normal ordering gives $H_I = E^{(1)} + \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \} + (V_2)_N$, with

$$E^{(1)} = \sum_i u_{ii} + \frac{1}{2} \sum_{ij} \overline{v}_{ijij},$$

$$E^{(2)} = \sum_{ia} \frac{|\tilde{f}_{ai}|^2}{\varepsilon_i - \varepsilon_a} + \frac{1}{4} \sum_{ijab} \frac{|\overline{v}_{abij}|^2}{D_{ij}^{ab}},$$

$$|\Psi^{(1)}\rangle = \sum_{ia} \frac{\tilde{f}_{ai}}{\varepsilon_i - \varepsilon_a} |\Phi_i^a\rangle + \frac{1}{4} \sum_{ijab} \frac{\overline{v}_{abij}}{D_{ij}^{ab}} |\Phi_{ij}^{ab}\rangle.$$

Here $D_{ij}^{ab} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b$. The special choice $h_0 = h$ gives $u = 0$ and $\tilde{f}_{ai} = U_{ai}$. Merely diagonalising h_0 does not imply $u = 0$.

Pause to think #1

(about 3 min – discuss with your neighbour)

Look at the list of fourth-order terms.

1. How many distinct terms does $E^{(4)}$ have when the chosen perturbation has $E^{(1)} = \langle H_I \rangle_0 = 0$?
2. In the third order, $E^{(1)} \langle VR^2V \rangle_0$ is the product of two quantities that are each sums over the whole system. If the system were two non-interacting copies of a smaller one, how would this product scale with the number of copies? And $E^{(3)}$ itself?

Answer to pause #1

1. Two: the principal term $\langle VRVRVRV \rangle_0$ and the folded term $-E^{(2)}\langle VR^2V \rangle_0$. For the Lipkin model ($E^{(1)} = 0$) `mbpt.py` gives -0.0036 and $+0.0072$, total $+0.0036$.
2. $E^{(1)}$ and $\langle VR^2V \rangle_0$ each scale linearly with the number of copies, so their product scales *quadratically* – an energy cannot. The principal term $\langle VRVRV \rangle_0$ contains a quadratically scaling piece of its own, and the subtraction removes it exactly: $E^{(3)}$ scales linearly. This is size extensivity, verified to machine precision in chapter 9, and on Friday it becomes the statement that only *linked* diagrams contribute.

Separating the reference scalar from the perturbation

Introduce $H(\lambda) = H_0 + \lambda H_I$ and intermediate normalisation $\langle \Phi_0 | \Psi_0(\lambda) \rangle = 1$. Define

$$E^{(1)} = \langle H_I \rangle_0, \qquad W = H_I - E^{(1)} I, \qquad \langle W \rangle_0 = 0.$$

Normal ordering relative to the determinant gives this separation explicitly:

$$W = \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \} + \frac{1}{4} \sum_{pqrs} \overline{v}_{pqrs} \{ a_p^\dagger a_q^\dagger a_s a_r \}, \qquad \tilde{f}_{pq} = u_{pq} + U_{pq} = f_{pq} - \varepsilon_p \delta_{pq}.$$

Here $u = h - h_0$ and $U_{pq} = \sum_j \overline{v}_{pjqj}$. The scalar is part of the total energy:

$$\boxed{E_0(\lambda) = \varepsilon_0 + \lambda E^{(1)} + \sum_{n \geq 2} \lambda^n E^{(n)}.}$$

Subtracting $\lambda E^{(1)} I$ shifts every eigenvalue by the same amount. The eigenvectors and all energy coefficients of order $n \geq 2$ are preserved. This scalar identity holds for any perturbation.

The scalar cancellation in the RS recursion

For $n \geq 2$, isolate the $k = 1$ term in the original recursion:

$$|\Psi^{(n)}\rangle = R_0 \left[(H_I - E^{(1)} I)|\Psi^{(n-1)}\rangle - \sum_{k=2}^{n-1} E^{(k)} |\Psi^{(n-k)}\rangle \right]$$
$$= R_0 \left[W|\Psi^{(n-1)}\rangle - \sum_{k=2}^{n-1} E^{(k)} |\Psi^{(n-k)}\rangle \right].$$

At first order, $R_0|\Phi_0\rangle = 0$ gives $|\Psi^{(1)}\rangle = R_0 H_I |\Phi_0\rangle = R_0 W|\Phi_0\rangle$. For $n \geq 1$,

$$E^{(n+1)} = \langle \Phi_0 | H_I | \Psi^{(n)} \rangle = \langle \Phi_0 | W | \Psi^{(n)} \rangle + E^{(1)} \underbrace{\langle \Phi_0 | \Psi^{(n)} \rangle}_{0}.$$

Perturbative order

The subtraction proportional to $E^{(1)}$ first occurs in $\Psi^{(2)}$ and contributes to the energy at third order. Using W absorbs these explicit scalar insertions into its matrix elements. Higher-order folded terms remain.

The third-order subtraction is contained in the middle vertex

Put $V = H_I$, $V_{mn} = \langle \Phi_m | V | \Phi_n \rangle$, and $D_m = \varepsilon_0 - \varepsilon_m$. Then

$$W_{mn} = V_{mn} - V_{00}\delta_{mn}, \qquad W_{m0} = V_{m0} \quad (m \neq 0).$$

Since $|\Psi^{(2)}\rangle = R_0 W R_0 W |\Phi_0\rangle$, its energy contribution is

$$E^{(3)} = \langle W R_0 W R_0 W \rangle_0 = \sum_{m,n\neq 0} \frac{V_{0m}(V_{mn} - V_{00}\delta_{mn})V_{n0}}{D_m D_n}$$

$$= \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{D_m D_n} - V_{00} \sum_{m\neq 0} \frac{|V_{m0}|^2}{D_m^2}.$$

The last term is exactly the original RS renormalisation term. The shorter W expression includes its effect. In particular, $\langle W \rangle_0 = 0$ permits nonzero excited-state diagonal elements $W_{mm} = V_{mm} - V_{00}$.

The folded term that survives at fourth order

The first three wave-function corrections in the shifted perturbation are

$$|\Psi^{(1)}\rangle = R_0 W|\Phi_0\rangle,$$
$$|\Psi^{(2)}\rangle = R_0 W R_0 W|\Phi_0\rangle,$$
$$|\Psi^{(3)}\rangle = R_0 W R_0 W R_0 W|\Phi_0\rangle - E^{(2)} R_0^2 W|\Phi_0\rangle.$$

Project with $\langle\Phi_0|W$:

$$\boxed{E^{(4)} = \langle W R_0 W R_0 W R_0 W\rangle_0 - E^{(2)}\langle W R_0^2 W\rangle_0.}$$

Since W and R_0 are Hermitian,

$$\langle W R_0^2 W\rangle_0 = \langle\Psi^{(1)}|\Psi^{(1)}\rangle = \sum_{m\neq 0} \frac{|V_{m0}|^2}{D_m^2} \geq 0.$$

Thus the surviving folded term is $-E^{(2)}\|\Psi^{(1)}\|^2$. For a gapped ground-state reference, $E^{(2)} \leq 0$, so this contribution is nonnegative. It is required even when the reference expectation of the perturbation is zero.

A two-level check with a nonzero first-order shift

Take a real perturbation and $\Delta > 0$:

$$H_0 = \begin{pmatrix} 0 & 0 \\ 0 & \Delta \end{pmatrix}, \qquad V = \begin{pmatrix} c & t \\ t & d \end{pmatrix}, \qquad W = V - cI = \begin{pmatrix} 0 & t \\ t & d-c \end{pmatrix}.$$

Direct expansion of the lower eigenvalue of $H_0 + \lambda V$ gives

$$E_-(\lambda) = \lambda c - \lambda^2 \frac{t^2}{\Delta} + \lambda^3 \frac{t^2(d-c)}{\Delta^2} + O(\lambda^4).$$

Using the original perturbation, the third-order terms are

$$\langle VR_0VR_0V\rangle_0 = \frac{t^2 d}{\Delta^2}, \qquad -E^{(1)}\langle VR_0^2V\rangle_0 = -\frac{ct^2}{\Delta^2}.$$

Using W, the middle matrix element is $d-c$ and the single expression gives their sum. The first-order energy is still c in the original partition.

Hartree–Fock and the choice of reference energy

For $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$ with canonical HF orbital energies,

$$\varepsilon_0 = \sum_i \varepsilon_i = \sum_i h_{ii} + \sum_{ij} \overline{v}_{ijij}, \qquad E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij} \overline{v}_{ijij}.$$

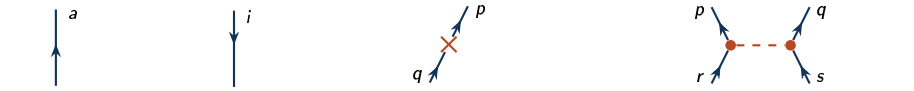
Consequently, in this partition

$$E^{(1)} = E_{\mathrm{HF}} - \varepsilon_0 = -\frac{1}{2}\sum_{ij} \overline{v}_{ijij}, \qquad \tilde{f}_{pq} = 0.$$

One can instead choose $H_0' = E_{\mathrm{HF}} + \sum_p \varepsilon_p \{a_p^\dagger a_p\} = H_0 + E^{(1)} I$ and perturb with W. Its zeroth-order energy is E_{HF} and its first-order correction is zero. The energy differences in R_0 and the coefficients at order two and higher agree in both partitions.

See Shavitt and Bartlett, *Many-Body Methods in Chemistry and Physics*, §2.4.4 and §3.6, for the shifted perturbation and normal ordering.

The elements of a diagram (week 38, chapter 5)



particle line | hole line | $\tilde{f}_{pq}$ (out p, in q) | $\overline{v}_{pqrs}$: ends (p,r) and (q,s)

- Time runs upward; the reference $|\Phi_0\rangle$ is the vacuum and is not drawn. Particle line: arrow up; hole line: arrow down.
- Every vertex is a product of *ends*, an end being $a^\dagger_{\rm out}\, a_{\rm in}$: the cross is one end, the interaction two ends joined by the dashed line, with the pairing (p,r), (q,s) of the matrix element.
- New this week: between successive vertices sits R_0; a horizontal *cut* between them will give the denominator.

These are antisymmetrised Goldstone (ASG) diagrams. A two-body vertex, drawn as two ends and a dashed line, carries an antisymmetrised matrix element. Its exchange partners are already included.

A term of order n is a string, read from the bottom

$$\langle WR_0WR_0\cdots R_0W\rangle_0 = \langle\Phi_0|W_nR_0W_{n-1}R_0\cdots R_0W_1|\Phi_0\rangle,$$

the subscript numbering the vertices from the one that acts first (rightmost, bottom) to the one that acts last (leftmost, top).

Forget the resolvents for a moment

Each W_k is a sum of normal-ordered monomials. The generalised Wick theorem (week 37, section 3.13): the vacuum expectation value of a product of normal-ordered groups is the sum over all *complete* contractions *between different groups*. Only two contractions are non-zero relative to $|\Phi_0\rangle$:

$$\boxed{\langle\Phi_0|a_a a_a^\dagger|\Phi_0\rangle = 1\ (a > F), \qquad \langle\Phi_0|a_i^\dagger a_i|\Phi_0\rangle = 1\ (i \le F),}$$

and in both the operator on the *left* belongs to the *later* vertex.

- First: $a_a^\dagger$ earlier creates a particle, a_a later absorbs it – a particle line, arrow up. Second: a_i earlier empties an occupied orbital, $a_i^\dagger$ later fills it – a hole line, arrow down.
- A complete contraction = a drawing in which every operator of every vertex is the end of exactly one line to a *different* vertex. Sum over contractions = sum over drawings: rule 1.

Pause to think #2 (about 3 min – discuss with your neighbour)

With normal-ordered vertices no line can begin and end at the same vertex.

1. What happened to the *bubbles* of last week's second-order diagrams (b), (c), (d) – a hole line leaving and re-entering the same interaction?
2. Draw the second-order diagrams that remain with normal-ordered vertices. How many are there?

Answer to pause #2

1. A bubble is a contraction *within* one vertex, $\sum_j \overline{v}_{pjqj} = U_{pq}$. Normal ordering has already carried it out: it is the U_{pq} inside $\tilde{f}_{pq} = u_{pq} + U_{pq}$. The cross now stands for "cross plus bubble", and the diagrams (b), (c), (d) of week 44 are the expansion of a single drawing in which the two crosses carry $\tilde{f}$.
2. Two: two crosses joined by a particle line and a hole line, and the ring – the interaction creating two pairs and the interaction annihilating them.

$$E^{(2)} = \sum_{ia} \frac{\tilde{f}_{ia}\tilde{f}_{ai}}{\varepsilon_i - \varepsilon_a} + \frac{1}{4}\sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b},$$

and $\tilde{f}_{ai} = f_{ai}$ for $a \neq i$: the coherent singles sum of week 44 is one diagram.

Thursday, lecture 2: The four rules, proved

One Wick contraction gives one orbital sum

Write $n_p = 1$ for an occupied reference orbital and $n_p = 0$ for an empty one. The determinant contractions are

$$\langle a_p^\dagger a_q \rangle_0 = \delta_{pq} n_p, \qquad \langle a_q a_p^\dagger \rangle_0 = \delta_{pq}(1 - n_p).$$

For example, take one one-body vertex $F = \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \}$ in each of two successive positions in the operator string. Normal ordering leaves only the intervertex pairing

$$\langle \{ a_p^\dagger a_q \} \{ a_r^\dagger a_s \} \rangle_0 = \delta_{ps} n_p \, \delta_{qr} (1 - n_q).$$

The two Kronecker deltas identify the endpoints. The occupation factors restrict $p = s = i$ to holes and $q = r = a$ to particles. Thus $\langle FF \rangle_0 = \sum_{ia} \tilde{f}_{ia} \tilde{f}_{ai}$. Each internal line has one independently summed orbital label.

The reduced resolvent on a determinant

For a nondegenerate reference, let $Q = 1 - |\Phi_0\rangle\langle\Phi_0|$. Define the inverse on the range of Q:

$$R_0 = \sum_{m\neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{\varepsilon_0 - \varepsilon_m}, \qquad R_0|\Phi_0\rangle = 0.$$

If M is a normal-ordered monomial, $[H_0, M] = \omega_M M$, where ω_M is the sum of the created orbital energies minus the sum of the annihilated ones. By the product rule for commutators,

$$[H_0, M_k \cdots M_1] = \Big(\sum_{j=1}^{k} \omega_{M_j}\Big) M_k \cdots M_1.$$

Whenever $M_k \cdots M_1|\Phi_0\rangle$ is nonzero, it has unperturbed energy $\varepsilon_0 + \sum_j \omega_{M_j}$. If a repeated occupied or empty label makes this state vanish, the determinant path already contributes zero.

Reading the energy denominator at a cut

A contraction entirely below a horizontal cut contributes $+\varepsilon_p - \varepsilon_p = 0$. An upward particle line leaves a creation operator below the cut. A downward hole line leaves an annihilation operator below it. Hence

$$D_k = \varepsilon_0 - \varepsilon_{m_k} = \sum_{i \text{ crossing}} \varepsilon_i - \sum_{a \text{ crossing}} \varepsilon_a .$$

For $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$ this gives

$$R_0 |\Phi_{ij}^{ab}\rangle = \frac{|\Phi_{ij}^{ab}\rangle}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.$$

A closed, time-ordered term with n vertices has $n-1$ cuts. Multiply by $1/D_k$ at each cut. A cut with no crossing lines returns to the reference and Q kills the term. Two independent excitations can overlap in time and survive this test.

The denominator rule and Pauli-forbidden label assignments

The resolvent argument applies directly to a nonzero intermediate determinant. In the Wick expansion a single contraction instead assigns independent labels to all its lines. Some assignments represent a vanishing determinant. For instance,

$$(a_b^\dagger a_i)(a_a^\dagger a_i)|\Phi_0\rangle = -a_b^\dagger a_a^\dagger a_i^2|\Phi_0\rangle = 0.$$

For fixed operator labels, Wick's theorem cancels this zero among its different pairings. Their cut energies are the same because contracted operators wholly below a cut contribute opposite orbital energies.

Two valid forms of the calculation

In determinant sums, use only allowed states. In an unrestricted diagram expansion, retain all particle and hole labels and the complete exchange partners. A separate Pauli veto on individual diagrams can spoil the cancellations, especially after applying the linked-diagram theorem.

A return to the reference at an empty cut is removed by Q. Repeated labels require the fermionic cancellation just described.

Counting contractions for antisymmetrised vertices

Each two-body vertex carries $1/(2!\,2!)$ and has two outgoing and two incoming slots. Exchanging two slots on one side reverses both the Wick sign and the sign of $\overline{v}_{pqrs}$, leaving their product unchanged.

For a fixed ordering of the vertices

At n_V two-body vertices there are 4^{n_V} choices of slot labels. Two choices give the same pairing when they exchange parallel lines at *both* endpoints. With at most two incoming and two outgoing slots, each pair of equivalent internal lines supplies two such choices.

$$\#\{\text{pairings of this topology}\} = \frac{4^{n_V}}{2^{n_{\mathrm{eq}}}}, \qquad w = \frac{1}{4^{n_V}} \frac{4^{n_V}}{2^{n_{\mathrm{eq}}}} = 2^{-n_{\mathrm{eq}}}.$$

Equivalent lines join the same two full interaction vertices in the same arrow direction. The vertex order is fixed. Internal orbital labels are summed without ordering restrictions. Distinct time orderings are counted separately, with no additional vertex factorial. The projecting bra fixes external labels in an amplitude.

Four contractions at second order

Take the top vertex as $\overline{v}_{pqrs}$ and the bottom vertex as $\overline{v}_{abij}$, with i,j occupied and a,b empty. The two particle contractions and the two hole contractions give

$$(\delta_{ra}\delta_{sb} - \delta_{rb}\delta_{sa})(\delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}).$$

Contracting the top matrix element with both brackets gives $4\overline{v}_{ijab}$. Thus the two vertex factors and four Wick pairings give $(1/4)^2\,4 = 1/4$.

$$E^{(2)}_{\text{doubles}} = \frac{1}{4}\sum_{ijab}\frac{|\overline{v}_{abij}|^2}{D^{ab}_{ij}} = \sum_{i<j}\sum_{a<b}\frac{|\overline{v}_{abij}|^2}{D^{ab}_{ij}}, \qquad D^{ab}_{ij} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b.$$

Every doubly excited determinant occurs once in the ordered sums. This checks the diagram weight independently against the determinant expression.

The sign, 1: vertex ends and the permutation π

The term of a complete contraction carries the sign of the permutation that brings the contracted pairs next to each other, in the order in which they appear (section 3.13). Reorder inside each vertex first:

$$\{a_p^\dagger a_q^\dagger a_s a_r\} = \{(a_p^\dagger a_r)(a_q^\dagger a_s)\}$$

(moving a_r two places left costs two signs; no contraction terms inside a normal-ordered product). So every vertex is a product of *ends* $a^\dagger_{\mathrm{out}}\, a_{\mathrm{in}}$, a cross being a single end. Number the ends $1, \ldots, K$ along the string.

Lines join an "out" to an "in"

A particle line contracts an $a_a^\dagger$ with an a_a, a hole line an a_i with a $a_i^\dagger$: in either case one operator is an out and the other an in. The lines therefore define a permutation π of the ends: the out-operator of end k is contracted with the in-operator of end $\pi(k)$.

- Follow a line to the end where it arrives, continue with the line leaving that end, ... until you are back: a closed loop is a cycle of π. So l = number of cycles.
- The number of lines equals the number of ends: $K = h + p$ (hole lines plus particle lines).

The sign, 2: $(-1)^{K-l}(-1)^{p} = (-1)^{h+l}$

Step 1: bring every pair into the order (out, in)

Fix one Wick pairing. In the alternating list of out and in slots, permute the incoming slots so that each creator sits next to its assigned annihilator. Its parity is $\operatorname{sgn}\pi = (-1)^{K-l}$. A cycle of length r requires $r-1$ transpositions, so all cycles require $\sum_\gamma (r_\gamma - 1) = K - l$. This counts the permutation sign of the selected contraction.

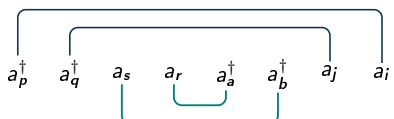
Step 2: compare with the order Wick's theorem wants (first in the string = first in the pair)

Hole line: $a_i^\dagger$ belongs to the later vertex and already stands left: value $\langle \Phi_0 | a_i^\dagger a_i | \Phi_0 \rangle = 1$, no extra sign.
Particle line: a_a belonged to the later vertex and originally stood first. Reverse that pair to its Wick order, contributing one extra minus sign and $\langle a_a a_a^\dagger \rangle_0 = 1$. Reordering complete two-operator pairs is even. Thus the orientation factor is $(-1)^p$.

$$\boxed{(-1)^{K-l}(-1)^{p} = (-1)^{h+p-l+p} = (-1)^{h+l}} \qquad \text{rule 4.}$$

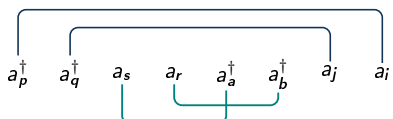
Exchanging two particle contractions changes one loop

Hold the hole contractions fixed in the second-order doubles diagram. In each string the four operators on the left belong to the upper vertex. Blue brackets fix $p = i$, $q = j$.



$r = a,\ s = b$

two loops, $h = 2$, Wick sign $+$



$r = b,\ s = a$

one loop, $h = 2$, Wick sign $-$

The exchange joins two cycles into one. At the same time the upper vertex changes from $\overline{v}_{ijab}$ to $\overline{v}_{ijba} = -\overline{v}_{ijab}$. The two signed terms therefore add. Exchanging the holes supplies the other two contractions counted above.

Open diagrams and the ordering of external lines

Fix $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$. To assign a sign to its amplitude, append the bra at the top of the diagram. Its operator string can be written

$$\langle \Phi_{ij}^{ab}| = \langle \Phi_0 | a_i^\dagger a_j^\dagger a_b a_a = \langle \Phi_0 | (a_i^\dagger a_a)(a_j^\dagger a_b).$$

Since a, b are virtual and i, j are occupied, moving a_a through the two middle operators produces two minus signs and no delta term.

- Pair (a, i) and (b, j) at the bra. Count all hole lines and cycles of this *closed* contraction, including those running to the bra. A loop closed this way is called a quasi-loop.
- Exchanging a and b in the external ordering reverses the determinant sign. When summing inequivalent attachments use the antisymmetriser $1 - P_{ab}$, where P_{ab} simply interchanges labels. The same convention applies to i, j.

The projecting bra carries unit weight. A wave-function amplitude with n physical vertices has an additional cut between the last vertex and the bra, for n denominators in total.

A fixed doubles amplitude and a summed wave function

For fixed distinct i, j, a, b and the stated determinant ordering, all four contractions with the projecting bra give

$$c_{ij}^{ab,(1)} = \langle \Phi_{ij}^{ab} | R_0 W | \Phi_0 \rangle = \frac{\overline{V}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.$$

To construct the wave function, sum independent determinants once:

$$|\Psi_{\text{doubles}}^{(1)}\rangle = \sum_{i<j,\, a<b} c_{ij}^{ab,(1)} |\Phi_{ij}^{ab}\rangle = \frac{1}{4} \sum_{ijab} c_{ij}^{ab,(1)} |\Phi_{ij}^{ab}\rangle.$$

Both the coefficient and determinant change sign when a, b or i, j are exchanged. Their product is invariant, giving four identical terms in the unrestricted sum. Thus a diagram for a fixed projected amplitude and a diagram for the full summed wave function have different external-label counting.

Pause to think #3

(about 4 min – discuss with your neighbour)

Apply the two proofs to the second-order ring (interaction creating two pairs, interaction annihilating them) and to the two-cross diagram.

1. List the ends and the permutation π for each. How many cycles? How many hole lines? What sign?
2. How many pairs of equivalent lines does each have, and what is the weight?
3. Which lines cross the single cut, and what is the denominator?

Answer to pause #3

1. Ring: ends $1L = (a, i)$, $1R = (b, j)$ (bottom), $2L = (i, a)$, $2R = (j, b)$ (top), writing (out, in). Lines: $1L \to 2L$ (a), $2L \to 1L$ (i): one cycle; $1R \to 2R$, $2R \to 1R$: a second. $l = 2$, $h = 2$: sign $+$. Two crosses: ends (a, i), (i, a), one 2-cycle, $l = 1$, $h = 1$: sign $+$.
2. Ring: the particle lines a, b both run from vertex 1 to vertex 2, the hole lines i, j both from 2 to 1 – two pairs, weight $\frac{1}{4}$. Two crosses: none, weight 1.
3. Ring: four lines cross, $1/(\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b)$. Two crosses: $1/(\varepsilon_i - \varepsilon_a)$. Together: $E^{(2)}$ of pause #2, now *derived* rather than checked.

The four rules for a closed, time-ordered diagram

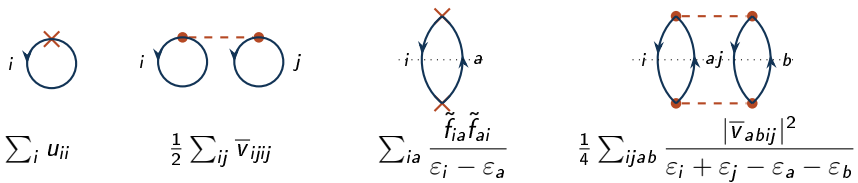
Use normal-ordered vertices, antisymmetrised two-body matrix elements and unrestricted internal orbital sums. For each distinct time ordering:

1. Assign one label per internal line. Sum holes over occupied orbitals and particles over empty orbitals. At every cross write $\tilde{f}_{pq}$, and at every interaction write $\overline{v}_{pqrs}$ with ends (p, r) and (q, s).
2. Insert $1/D_k$ between consecutive vertices, with $D_k = \sum_{\text{holes}} \varepsilon_i - \sum_{\text{particles}} \varepsilon_a$ on that cut.
3. Multiply by $2^{-n_{\text{eq}}}$ for the equivalent internal line pairs.
4. Multiply by $(-1)^{h+l}$ using the hole lines and closed fermion loops of the chosen vertex pairing.

Sign check

The Wick sign and the signs of the denominators are separate. The doubles diagram has $h = l = 2$, so its Wick sign is $+1$. For a ground-state reference $D_{ij}^{ab} < 0$, and its contribution to $E^{(2)}$ is negative whenever the matrix element is nonzero.

First and second order, re-read with the rules



$\sum_i u_{ii}$ $\quad \frac{1}{2}\sum_{ij} \overline{v}_{ijij}$ $\quad \sum_{ia} \frac{\tilde{f}_{ia}\tilde{f}_{ai}}{\varepsilon_i - \varepsilon_a}$ $\quad \frac{1}{4}\sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$

- First order uses the bare vertices. The two-body expectation is $\frac{1}{2}\sum_{ij} \overline{v}_{ijij}$ from two Wick pairings times $1/4$. Both diagrams have positive Wick sign and no cut. Together they give $E_{\mathrm{ref}} - \varepsilon_0$.
- Second order (normal-ordered vertices): the two diagrams of pause #3.
- Hartree-Fock basis: f is diagonal with eigenvalues ε_p, so $\tilde{f}_{pq} = 0$ for all p, q: the bare one-body residual u and the contracted mean field U cancel in $\tilde{f}$. Every effective cross vanishes and only the doubles diagram contributes to MP2. Pairing model: $\tilde{f}_{ai} = 0$ in the pairing basis, ring $= -\frac{7}{24}g^2$.

Friday, lecture 3: Third order, enumerated

Enumerating the third order by intermediate states

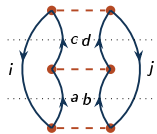
$E^{(3)} = \langle WR_0WR_0W\rangle_0$: all connected drawings with three normal-ordered vertices. The intermediate states can only be singles (S) or doubles (D):

- Bottom vertex on the reference: a cross creates a single ($\tilde{f}_{ai}$), an interaction a double ($\overline{v}_{abij}$) – a normal-ordered operator has to *create* every line it touches.
- Top vertex: a cross closes a single ($\tilde{f}_{ia}$), an interaction a double ($\overline{v}_{ijab}$).
- Middle vertex: S→S, S→D, D→S, D→D, by a cross or by an interaction. Triples and quadruples cannot be closed by the top vertex; a return to the reference is killed by Q.

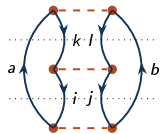
	vertices (bottom to top)	states	description	sign, weight
1, 2	cross, cross, cross	S, S	particle-line insertion / hole-line insertion	$+$ / $-$
3	cross, cross, interaction	S, D	two pairs created by crosses	$+$
4	cross, interaction, cross	S, S	particle-hole scattering	$-$
5, 6	cross, interaction, interaction	S, D	particle line / hole line into the interaction	$+\frac{1}{2}$ / $-\frac{1}{2}$
7, 8	interaction, cross, interaction	D, D	particle-line / hole-line insertion	$+\frac{1}{2}$ / $-\frac{1}{2}$
9	interaction, cross, cross	D, S	two pairs closed by crosses	$+$
10, 11	interaction, interaction, interaction	D, D	particle-particle ladder / hole-hole ladder	$+\frac{1}{8}$ / $+\frac{1}{8}$
12	interaction, interaction, interaction	D, D	particle-hole ring	$-$
13, 14	interaction, interaction, cross	D, S	particle lines / hole lines into the interaction	$+\frac{1}{2}$ / $-\frac{1}{2}$

Fourteen diagrams. Nothing forgotten, nothing counted twice: that is what the pictures are for.

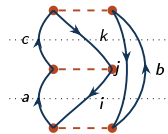
The three diagrams with three interactions: MP3



(10)



(11)



(12)

$$\Delta E^{(3)}_{\mathrm{MP}} = \frac{1}{8}\sum \frac{\overline{v}_{ijcd}\overline{v}_{cdab}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{cd}_{ij}} + \frac{1}{8}\sum \frac{\overline{v}_{klab}\overline{v}_{ijkl}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{ab}_{kl}} + \sum \frac{\overline{v}_{kjcb}\overline{v}_{icak}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{cb}_{kj}},$$

with $\varepsilon^{ab}_{ij} \equiv \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b$.

Ladders: three pairs of equivalent lines, $\frac{1}{8}$; $h = 2$ or 4, $l = 2$: $+$. Ring: no equivalent lines, $h = 3$, $l = 2$: $-$ in front of $\overline{v}_{ciak}$, i.e. $+$ with $\overline{v}_{icak}$.

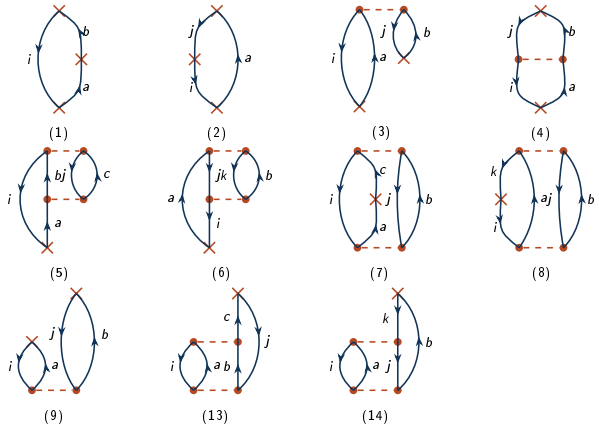
MP3: what the three diagrams say

- In a canonical Hartree-Fock basis $\tilde{f}_{pq} = f_{pq} - \varepsilon_p \delta_{pq} = 0$: every diagram with a cross vanishes, and the three are the *only* third-order diagrams – the Møller-Plesset third-order energy.
- Particle-particle ladder: the two excited particles scatter off each other once before they fall back. Hole-hole ladder: the same for the two holes. Ring: one particle-hole pair scatters into another.
- Each is the third-order member of an infinite family. Iterating the pp channel with a Pauli-projected two-particle propagator gives Brueckner's G-matrix. Rings of every length: the RPA of chapter 7 (week 40, and the stability matrix of week 39). Both at once, with the terms that mix them: coupled-cluster doubles, chapter 10.

Many-body connection

The linked-diagram theorem (week 46) removes unlinked contributions to the energy. It motivates organising approximations by linked diagram families; it does not by itself establish the convergence or accuracy of a selected resummation.

The eleven diagrams with at least one cross



All vanish in a canonical HF basis. In the models of chapter 9 (H_0 diagonal by construction, reference not HF) the cross is the bubble U_{pq}: 7 and 8 are the first self-energy insertions on a particle and a hole line. Mirror images in time: 3/9, 5/13, 6/14.

The full third order, read off the drawings: diagrams 1–7

$$
\begin{aligned}
E^{(3)} = &\sum_{iab} \frac{\tilde{f}_{ib}\tilde{f}_{ba}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_i-\varepsilon_b)} - \sum_{ija} \frac{\tilde{f}_{ja}\tilde{f}_{ij}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_j-\varepsilon_a)} + \sum_{ijab} \frac{\overline{v}_{ijab}\tilde{f}_{bj}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{ij}^{ab}} \\
&- \sum_{ijab} \frac{\tilde{f}_{jb}\overline{v}_{ibja}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_j-\varepsilon_b)} + \frac{1}{2}\sum \frac{\overline{v}_{ijbc}\overline{v}_{bcaj}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{ij}^{bc}} - \frac{1}{2}\sum \frac{\overline{v}_{jkab}\overline{v}_{ibjk}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{jk}^{ab}} \\
&+ \frac{1}{2}\sum \frac{\overline{v}_{ijcb}\tilde{f}_{ca}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\,\varepsilon_{ij}^{cb}} + \cdots \qquad (\varepsilon_{ij}^{ab} = \varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b)
\end{aligned}
$$

A sign worth checking: diagram 2, the cross on a hole line

Ends $(a,i),(i,j),(j,a)$: π is one 3-cycle, $l=1$; $h=2$: sign $-$. The cross adds a line, not a loop; whether the added line is a hole or a particle decides the sign – the $f_{ab}C_i^b$ versus $-f_{ji}C_j^a$ of the singles equation of week 38. Diagram 1 (cross on the particle line) has the same single cycle but $h=1$: $+$.

Friday, lecture 4: Third order, evaluated

The full third order, read off the drawings: diagrams 8–14

$$
\begin{aligned}
\cdots &- \frac{1}{2}\sum \frac{\overline{v}_{kjab}\tilde{f}_{ik}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kj}^{ab}} + \sum_{ijab} \frac{\tilde{f}_{jb}\tilde{f}_{ia}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_j - \varepsilon_b)} + \frac{1}{8}\sum \frac{\overline{v}_{ijcd}\overline{v}_{cdab}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{ij}^{cd}} \\
&+ \frac{1}{8}\sum \frac{\overline{v}_{klab}\overline{v}_{ijkl}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kl}^{ab}} - \sum \frac{\overline{v}_{kjcb}\overline{v}_{ciak}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kj}^{cb}} \\
&+ \frac{1}{2}\sum \frac{\tilde{f}_{jc}\overline{v}_{icab}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_j - \varepsilon_c)} - \frac{1}{2}\sum \frac{\tilde{f}_{kb}\overline{v}_{ijak}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_k - \varepsilon_b)}.
\end{aligned}
$$

A sign worth checking: diagram 12, the ring

Ends $1L{=}(a,i)$, $1R{=}(b,j)$, $2L{=}(c,a)$, $2R{=}(i,k)$, $3L{=}(k,c)$, $3R{=}(j,b)$. Lines: $1L \to 2L$ (a), $2L \to 3L$ (c), $3L \to 2R$ (k), $2R \to 1L$ (i) – one cycle; $1R \to 3R$ (b), $3R \to 1R$ (j) – a second. $l = 2$, $h = 3$: $-$ in front of $\overline{v}_{ciak}$, which is $+$ in front of $\overline{v}_{icak}$, the form usually quoted.

Pause to think #4

(about 4 min – discuss with your neighbour)

The pairing model: the interaction only moves a pair $(p\uparrow, p\downarrow) \rightarrow (q\uparrow, q\downarrow)$, H_0 is diagonal by construction, $u = 0$.

1. Which of the fourteen diagrams can be non-zero? (Think: can the interaction connect the reference to a single? Can it scatter a particle and a hole from different pairs? What is $\tilde{f}_{pq} = U_{pq} = \sum_j \overline{v}_{pjqj}$ for an occupied and for an empty spin-orbital?)
2. Week 44 found principal -0.02300347, renormalisation -0.01258681, $E^{(3)} = -0.01041667$ at $g = 0.5$. Guess which diagrams account for which number.

Answer to pause #4

1. $\tilde{f}_{ai} = 0$ (a pair cannot be half moved) kills 1–6, 9, 13, 14; the ring needs a particle-hole scattering between different pairs and vanishes; $U_{pp} = \overline{v}_{p\bar{p}p\bar{p}} = -g/2$ for an *occupied* spin-orbital, 0 for an empty one. Left: the two ladders and the hole insertion 8.
2. Four levels, two pairs, unit spacing: doubles $i \to a$ with $D_{ia} = 2(\varepsilon_i - \varepsilon_a) = -4, -6, -2, -4$, element $-g/2$.

$$\text{pp} = -\tfrac{g^3}{8}\sum_i\Big(\sum_a \tfrac{1}{D_{ia}}\Big)^2 = -\tfrac{53}{576}g^3, \qquad \text{hh} = -\tfrac{53}{576}g^3,$$

$$\text{insertion} = 2\cdot\tfrac{g}{2}\sum_{ia}\tfrac{(g/2)^2}{D_{ia}^2} = +\tfrac{29}{288}g^3, \qquad \boxed{E^{(3)} = -\tfrac{1}{12}g^3.}$$

At $g = 0.5$: $-0.0115017 - 0.0115017 + 0.0125868 = -0.0104167$. The ladders *are* the principal term; the insertion *is* minus the renormalisation term.

Where the "correction larger than the answer" went

In the bookkeeping of week 44

The diagonal element V_{mm} of the middle vertex contains the reference energy V_{00}; the renormalisation term subtracts it. What is left, $W_{mm} = V_{mm} - V_{00}$, is the diagonal part of the ladders plus the mean-field insertions on the four lines of the double.

Pairing model

For a pair excitation the two cancel: $W_{mm} = 0$. So the hole insertion, $+g \sum_m |V_{0m}|^2/D_m^2$ with $-\tilde{f}_{ii} = g/2$ on each hole line, equals $-V_{00} \sum_m |V_{0m}|^2/D_m^2$ exactly, and the principal term is the ladders alone.

Measured

`diagrams.py`, pairing $g = 0.5$:
pp ladder -0.011501736
hh ladder -0.011501736
hole insertion $+0.012586806$
sum $-0.010416667 = E^{(3)}$

Pairing + particle-hole, $g = 0.5$, $f = 0.025$: ladders -0.0153905 each, insertion $+0.0169196$, ring 0, diagrams 2, 3, 5, 6, 9, 13, 14 switched on at $\leq 1.5 \times 10^{-4}$ ($O(f^2)$: two vertices only the particle-hole term supplies); sum $-0.0138615 = E^{(3)}$.

The renormalisation term was never a correlation correction: it removes a mean-field contribution. (The two ladders are equal because the model has as many empty levels as filled ones at the same spacing.)

Reading a diagram into code: `diagrams.py`

With `eps`, `ftil`, `vbar` and the occupied/empty index lists `o`, `v`; `D1[i,a]` $= \varepsilon_i - \varepsilon_a$, `D2[i,j,a,b]` $= \varepsilon^{ab}_{ij}$:

```
v_oovv = vb[np.ix_(o, o, v, v)] # <ij|v|ab> v_vvoo = vb[np.ix_(v, v, o, o)] # <ab|v|ij>
v_vvvv = vb[np.ix_(v, v, v, v)] # <cd|v|ab> v_oooo = vb[np.ix_(o, o, o, o)] # <ij|v|kl>
v_vovo = vb[np.ix_(v, o, v, o)] # <ci|v|ak> foo, fov, fvo, fvv = blocks of ftil
# 10 particle-particle ladder: three equivalent pairs -> 1/8, h = 2, l = 2 -> +
pp = 0.125 * np.einsum("ijcd,cdab,abij,ijab,ijcd->", v_oovv, v_vvvv, v_vvoo, 1/D2, 1/D2)
# 11 hole-hole ladder
hh = 0.125 * np.einsum("klab,ijkl,abij,ijab,klab->", v_oovv, v_oooo, v_vvoo, 1/D2, 1/D2)
# 12 particle-hole ring: no equivalent lines, h = 3, l = 2 -> -
ring = -np.einsum("kjcb,ciak,abij,ijab,kjcb->", v_oovv, v_vovo, v_vvoo, 1/D2, 1/D2)
# 8 hole-line insertion on a double: one equivalent pair (a,b) -> 1/2, h = 3, l = 2 -> -
ins = -0.5 * np.einsum("kjab,ik,abij,ijab,kjab->", v_oovv, foo, v_vvoo, 1/D2, 1/D2)
# 2 cross on a hole line: h = 2, l = 1 -> -
d2 = -np.einsum("ja,ij,ai,ia,ja->", fov, foo, fvo, 1/D1, 1/D1)
```

Each line is one diagram: the matrix elements in the order top to bottom, one $1/D$ per cut, the weight and the sign in front. `GeneralHamiltonian` builds the same Hamiltonian in the determinant space and runs the recursion of week 44, so the two routes can be compared for any $(\varepsilon, u, \bar{v})$; the fourteen diagrams and the determinant-space $E^{(3)}$ agree to 10^{-12} for every Hamiltonian tried. The notebook `week45.ipynb` is self-contained and repeats every number on these frames.

Which pair ladders are we summing?

The question. Promote one pair, scatter it within the four $2p2h$ determinants, then return to the reference. What do these *unfolded chains* sum to when the energy in every propagator is fixed?

The pair space (four levels, two pairs, $\xi = 1$). The interaction moves pairs, so such a term only visits the four determinants $|\Phi_i^a\rangle$, $i \in \{1,2\}$ emptied, $a \in \{3,4\}$ filled, with

$$D_i^a = \varepsilon_0 - \varepsilon_i^a = -2(a-i), \quad \text{i.e. } (-2,-4,-4,-6),$$

all coupled to the reference by $w_m = \langle \Phi_m | W | \Phi_0 \rangle = \overline{v}_{i\bar{i}a\bar{a}} = -g/2$.

The rungs. $W_{mm} = V_{mm} - V_{00} = -g - (-g) = 0$: the diagonal ladder parts cancel the hole insertions (Friday's result). Off the diagonal, $\langle \Phi_i^b | W | \Phi_i^a \rangle = -g/2$ is a *pp rung*, $\langle \Phi_j^a | W | \Phi_i^a \rangle = -g/2$ an *hh rung*, and $\langle \Phi_j^b | W | \Phi_i^a \rangle = 0$ (two pairs would have to move).

In the order $(\Phi_2^3, \Phi_1^3, \Phi_2^4, \Phi_1^4)$:

$$W_{DD} = -\frac{g}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix},$$

and $R_0 = \mathrm{diag}(1/D_m)$. The four states form a *square*: each is a neighbour of the two that share one of its indices.

At order $n \geq 2$ the unfolded chain contributes

$$w^T R_0 \left(W_{DD} R_0 \right)^{n-2} w,$$

with $n-2$ internal rungs and $n-1$ propagators. Energy feedback produces folded terms starting at fourth order.

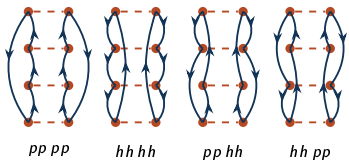
Unfolded pair chains through fourth order

$$E^{(2)} = \sum_m \frac{w_m^2}{D_m} = -\frac{g^2}{4}\left(\tfrac{1}{2}+\tfrac{1}{4}+\tfrac{1}{4}+\tfrac{1}{6}\right) = -\frac{7}{24}g^2,$$

$$E^{(3)} = \sum_{m\neq n} \frac{w_m W_{mn} w_n}{D_m D_n} = -\frac{g^3}{4}\left(\tfrac{1}{8}+\tfrac{1}{8}+\tfrac{1}{24}+\tfrac{1}{24}\right)$$
$$= -\frac{1}{12}g^3$$

(at third order *every* intermediate state is $2p2h$: this is Friday's $-\frac{53}{576} - \frac{53}{576} + \frac{29}{288}$, the diagonal parts already cancelled).

$$E^{(4)}_{2p2h} = \sum_{m,n,k} \frac{w_m W_{mn} W_{nk} w_k}{D_m D_n D_k} = -\frac{7}{288}g^4,$$



The two internal rungs have four pp/hh orderings. Every cut between vertices crosses two particle lines and two hole lines. The weights on the left use the net matrix elements of W_{DD} after the diagonal cancellation.

There are $4 \cdot 2^2 = 16$ walks: four starting determinants and two choices at each of two steps. Each walk visits three determinants, with repetitions allowed. For example, $\Phi_2^3 \to \Phi_1^3 \to \Phi_2^3$ gives $(g/2)^4/((-2)(-4)(-2)) = -g^4/256$.

What the restriction leaves out at fourth order

$$E^{(4)} = \langle W R_0 W R_0 W R_0 W \rangle_0 - E^{(2)} \langle W R_0^2 W \rangle_0$$

Through the $4p4h$ state. Both pairs lifted, $D = -8$, coupled by $-g/2$ to each of the four pair states:

$$E^{(4)}_{4p4h} = \frac{g^4}{16} \Big(\sum_m \frac{1}{D_m} \Big)^2 \frac{1}{-8} = -\frac{49}{4608} g^4 .$$

The folded term.

$$-E^{(2)} \sum_m \frac{w_m^2}{D_m^2} = \frac{7}{24} g^2 \cdot \frac{g^2}{4} \cdot \frac{29}{72} = +\frac{203}{6912} g^4 .$$

Measured

$g = 0.5$:
$2p2h$ ladders -0.00151910
$4p4h$ -0.00066461
folded $+0.00183558$
sum $-0.00034813 = E^{(4)}$ (week-44 recursion)

$$E^{(4)} = \Big(-\tfrac{336}{13824} - \tfrac{147}{13824} + \tfrac{406}{13824} \Big) g^4 = -\frac{77}{13824} g^4 .$$

The $4p4h$ and folded terms partially cancel. Together they add $259 g^4/13824$ to the unfolded chain. The chain alone has $336/77 = 48/11 \simeq 4.36$ times the magnitude of the full fourth-order correction.

A Bethe–Salpeter-type ladder equation in the restricted pair space

From chains to pair amplitudes. Write $t^{(1)} = R_0 w$. Associativity gives

$$R_0 (W_{DD} R_0)^k = (R_0 W_{DD})^k R_0 .$$

Hence the kernel acting on an amplitude is $K = R_0 W_{DD}$. Summing the iterates gives

$$t = (I - K)^{-1} t^{(1)}, \qquad \Delta E_{\mathrm{lad}} = w^T t .$$

The amplitude satisfies

$$\boxed{t_m = \frac{w_m + \sum_n W_{mn} t_n}{D_m}}.$$

Since $E_{\mathrm{ref}} - H_{mm} = D_m - W_{mm}$,

$$\Delta E_{\mathrm{lad}} = H_{0D} (E_{\mathrm{ref}} I - H_{DD})^{-1} H_{D0}.$$

Why every second order repeats. In the ordering of the four pair determinants,

$$t^{(1)} = g(\tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{8}, \tfrac{1}{12})^T,$$
$$K t^{(1)} = g^2(\tfrac{1}{16}, \tfrac{1}{24}, \tfrac{1}{24}, \tfrac{1}{48})^T,$$
$$K^2 t^{(1)} = \tfrac{g^2}{12}\, t^{(1)}.$$

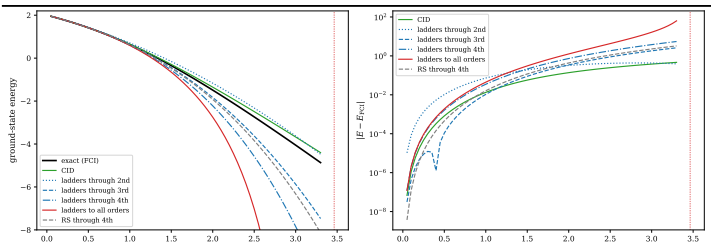
Thus $E_{\mathrm{lad}}^{(n+2)} = (g^2/12) E_{\mathrm{lad}}^{(n)}$. The even and odd subsequences are separately geometric. Adding them gives

$$\boxed{\Delta E_{\mathrm{lad}} = \frac{-7g^2/24 - g^3/12}{1 - g^2/12} = -\frac{g^2(7 + 2g)}{2(12 - g^2)}}.$$

The series converges for $|g| < 2\sqrt{3}$. Its fourth- and fifth-order terms are $-7g^4/288$ and $-g^5/144$.

Pair ladders compared with CID, CCD and FCI

g	pp only	pp+hh chains	CID	CCD	FCI
0.25	1.731092	1.730366	1.730507	1.730456	1.730460
0.50	1.421404	1.414894	1.417596	1.416638	1.416774
1.00	0.658404	0.590909	0.648907	0.630443	0.635548
1.50	−0.343227	−0.653846	−0.291132	−0.384165	−0.348191
2.00	−1.664596	−2.750000	−1.352087	−1.609594	−1.489652



All entries are total energies ($E_{\mathrm{ref}} = 2 - g$). The pp-only column keeps the off-diagonal pp moves of W_{DD} with the same fixed denominators. The plotted ladder curves include pp and hh moves; the red dotted line marks their pole at $2\sqrt{3}$.

Frozen denominators, CID and the convergence radius

Restoring energy feedback. In the same reference-plus-doubles space, eliminating the doubles gives the CID ground-state equation

$$\Delta E_{\mathrm{CID}} = H_{0D}\big[(E_{\mathrm{ref}} + \Delta E_{\mathrm{CID}})I - H_{DD}\big]^{-1} H_{D0}.$$

Freezing the energy argument at E_{ref} gives the pp+hh chain sum. Expanding the CID resolvent in ΔE_{CID} generates the folded terms. The first fold is $-E^{(2)} w^T R_0^2 w$.

Why the full series has a smaller radius. For $y = E + g - 6$ the ground-state branch of the full pairing Hamiltonian satisfies

$$p(y, g) = y^4 - (20 + 3g^2)y^2 + 2g^3 y + 64 + 20g^2 = 0, \qquad y(0) = -4.$$

The nearest coalescences on this branch satisfy $p = \partial_y p = 0$ at $g \simeq 0.679965 \pm 1.360006\, i$. Their modulus gives the full Rayleigh–Schrödinger radius $R_{\mathrm{RS}} \simeq 1.52052$. The frozen chain has poles at $g = \pm 2\sqrt{3}$, where $E_{\mathrm{ref}} I - H_{DD}$ is singular.

At $g = 2$ the chain series converges to $E = -2.750000$, while FCI gives -1.489652. Even at $g = 0.25$ its absolute error is 9.3×10^{-5}, compared with 8.9×10^{-6} for third order.

Pause to think #5

(about 3 min – discuss with your neighbour)

The restricted pair equation uses $R_0 = \mathrm{diag}(1/D_m)$.

1. Replace $1/D_m$ by $1/(D_m + \Delta E)$ and solve for ΔE self-consistently. Which eigenvalue problem does this solve? Which determinant would be needed for FCI?
2. A pp rung keeps the hole pair i fixed. For each i, the excited pair can occupy $a = 3$ or 4. Use the pp-only and FCI columns at $g = 1.5$ and $g = 2$ to test whether the pp-only energy provides an upper bound.

Answer to pause #5

1. This is the Brillouin–Wigner equation in the reference-plus-doubles space. Its ground-state root is CID. Including the determinant in which both pairs have been promoted extends this space to the full paired sector.
2. Put $v = -g/2$. For each fixed i, inverting the two-state pp block gives

$$\begin{pmatrix} D_i^3 & -v \\ -v & D_i^4 \end{pmatrix} \begin{pmatrix} t_i^3 \\ t_i^4 \end{pmatrix} = v \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \Delta E_{\mathrm{pp}} = \sum_{i=1}^{2} v^2 \, \frac{D_i^3 + D_i^4 + 2v}{D_i^3 D_i^4 - v^2}.$$

At $g = 1.5$ the pp energy lies above FCI by 0.004964. At $g = 2$ it lies below FCI by 0.174944. The fixed-energy resolvent therefore gives no variational upper bound.

Here *pp only* refers to the pp moves of the model's zero-diagonal W_{DD}. The conventional Brueckner G-matrix iterates a two-body interaction with a Pauli-projected two-particle propagator at the incoming pair's starting energy.

A look ahead at the fourth-order diagrams

$$E^{(4)} = \langle W R_0 W R_0 W R_0 W \rangle_0 - E^{(2)} \langle W R_0^2 W \rangle_0 .$$

Four normal-ordered vertices can form two closed components, each with two vertices. For disjoint excitations m, n:

vertex pairs	intermediate states	status
1–3 and 2–4	$m,\ m \otimes n,\ n$	unlinked, overlapping in time
1–4 and 2–3	$m,\ m \otimes n,\ m$	unlinked, overlapping in time
1–2 and 3–4	m, Φ_0, n	zero, the middle Q removes the reference

The folded term is also a product of two second-order quantities. Adding the overlapping time orderings permits its cancellation against the unlinked part of the principal term. Shared labels require the EPV exchange bookkeeping.

Week 46 develops the factorisation, the linked-diagram theorem and its proof, the wave operator, and the Pauli principle in intermediate states.

Summary

Summary of week 45

- The RS recursion generates the wave operator through third order and the energy through fourth order, including the folded terms.
- $W = H_I - E^{(1)} I$ absorbs the explicit first-order scalar insertions. $\varepsilon_0 + E^{(1)}$ remains in the reference energy. The fourth-order subtraction is $-E^{(2)} \|\Psi^{(1)}\|^2$.
- For time-ordered antisymmetrised diagrams, Wick contractions give the label sums. The cuts give the resolvent denominators. Equivalent internal line pairs give $1/2$ each, and the Wick sign is $(-1)^{h+l}$.
- Fourteen connected third-order diagrams occur for general one- and two-body normal-ordered vertices. With a canonical HF partition only the pp ladder, hh ladder and ph ring survive.
- The pairing model illustrates how the original third-order subtraction enters through the diagonal matrix elements of W. At fourth order, unlinked graphs require the factorisation developed next week.
- At fixed reference energy, the unfolded pp+hh chains through the four $2p2h$ determinants solve $t = R_0 w + R_0 W_{DD} t$. Even and odd orders are separately geometric with ratio $g^2/12$, giving $\Delta E_{\mathrm{lad}} = -g^2(7 + 2g)/[2(12 - g^2)]$. Restoring the energy dependence generates the CID folded terms.

Next week

Week 46, November 12–13. The linked-diagram theorem and coupled cluster

Thursday develops the wave operator and its linked and unlinked diagrams, fourth-order cancellation, denominator factorisation, the proof in imaginary time, and exclusion-principle-violating terms.

Friday covers coupled-cluster theory, including the exponential ansatz, the similarity-transformed Hamiltonian, termination of the expansion, the CCD equations and the pairing model.

Reading is chapter 9, section 9.15 from the unlinked diagrams onward, chapter 10, sections 10.1–10.7, and Shavitt and Bartlett, chapters 5–6 and 9. The week-45 exercises are discussed in the exercise session and answered in the week-46 slides.

Answers to the week-44 exercises

Exercise 1: the energy as a projection, and a)

$H = H_0 + V$, $H_0|\phi_n\rangle = \varepsilon_n|\phi_n\rangle$, $H|\psi_n\rangle = E_n|\psi_n\rangle$; ground states non-degenerate and $\langle\phi_0|\psi_0\rangle \neq 0$. Nothing is assumed about the norm of $|\psi_0\rangle$.

The projection

Project $H|\psi_0\rangle = E_0|\psi_0\rangle$ onto $\langle\phi_0|$; $\langle\phi_0|H_0 = \varepsilon_0\langle\phi_0|$ (Hermiticity), so

$$(E_0 - \varepsilon_0)\langle\phi_0|\psi_0\rangle = \langle\phi_0|V|\psi_0\rangle \quad \Longrightarrow \quad \boxed{E_0 - \varepsilon_0 = \frac{\langle\phi_0|V|\psi_0\rangle}{\langle\phi_0|\psi_0\rangle}.}$$

Exact; a transition element, not an expectation value, hence not variational. Non-degeneracy is not needed here – it is needed when $z - H_0$ is inverted on the Q space.

a) Idempotency

$P^2 = |\phi_0\rangle\langle\phi_0|\phi_0\rangle\langle\phi_0| = P$; $Q^2 = (1-P)^2 = 1 - 2P + P^2 = 1 - P = Q$. Also Hermitian, $PQ = QP = 0$, $P + Q = 1$, and $[P, H_0] = [Q, H_0] = 0$ because $|\phi_0\rangle$ is an eigenstate of H_0: $Q = \sum_{n\neq 0} |\phi_n\rangle\langle\phi_n|$, and any $f(H_0)$ is diagonal in the same basis, so $Qf(H_0)Q = Qf(H_0) = f(H_0)Q$ – the licence to write $Q/(z - H_0)$ (week 44, lecture 1).

Exercise 1 b): the expansion for arbitrary z

$|\psi_0\rangle = (P+Q)|\psi_0\rangle = \langle\phi_0|\psi_0\rangle|\phi_0\rangle + Q|\psi_0\rangle$. Add and subtract $z|\psi_0\rangle$ in the Schrödinger equation:

$$(z - H_0)|\psi_0\rangle = (z - E_0 + V)|\psi_0\rangle.$$

Multiply by Q; on the Q space $z - H_0$ has eigenvalues $z - \varepsilon_n \neq 0$ ($n \neq 0$) and can be inverted:

$$\frac{Q}{z - H_0} \equiv Q(z - H_0)^{-1} Q = \sum_{n\neq 0} \frac{|\phi_n\rangle\langle\phi_n|}{z - \varepsilon_n}, \qquad Q|\psi_0\rangle = \frac{Q}{z - H_0}(z - E_0 + V)|\psi_0\rangle.$$

(At $z = \varepsilon_0$ the full inverse has a pole; the reduced one exists iff the ground state of H_0 is non-degenerate – *this* is where the assumption enters.) Call $A(z) = \frac{Q}{z-H_0}(z - E_0 + V)$ and iterate:

$$|\psi_0\rangle = \langle\phi_0|\psi_0\rangle\big(1 + A + \cdots + A^N\big)|\phi_0\rangle + A^{N+1}|\psi_0\rangle \xrightarrow{N\to\infty} \boxed{|\psi_0\rangle = \langle\phi_0|\psi_0\rangle \sum_{n=0}^{\infty} \Big(\frac{Q}{z - H_0}(z - E_0 + V)\Big)^n |\phi_0\rangle,}$$

a formal Neumann series (convergence not guaranteed); the overlap is an overall factor, as linearity demands. Insert into the projected identity:

$$\boxed{E_0 = \varepsilon_0 + \sum_{n=0}^{\infty} \langle\phi_0|V\Big(\frac{Q}{z - H_0}(z - E_0 + V)\Big)^n |\phi_0\rangle.}$$

The overlap has cancelled – which is why one may set $\langle\phi_0|\psi_0\rangle = 1$ from the start. Still implicit: E_0 sits inside $z - E_0 + V$ for every z but one.

Exercise 1 c): the two schemes

$z = E_0$: Brillouin-Wigner

$z - E_0 + V = V$: only interactions and resolvents alternate,

$$E_0 = \varepsilon_0 + V_{00} + \sum_{m\neq 0} \frac{V_{0m}V_{m0}}{E_0 - \varepsilon_m} + \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{(E_0 - \varepsilon_m)(E_0 - \varepsilon_n)} + \cdots$$

Explicit and simple, but E_0 in every denominator: implicit, self-consistent; a truncation after k vertices is *not* a polynomial of degree k in the coupling. The full resummation (Feshbach partitioning, week 44) is exact. Finite self-consistent truncations can violate size extensivity.

$z = \varepsilon_0$: Rayleigh-Schrödinger

$R_0 = Q/(\varepsilon_0 - H_0)$ is known, but $z - E_0 + V = V - \Delta E$ and ΔE must be expanded too ($R_0 \Delta E|\phi_0\rangle = 0$ removes the rightmost one); with $D_m = \varepsilon_0 - \varepsilon_m$,

$$E^{(1)} = V_{00}, \qquad E^{(2)} = \sum_{m\neq 0} \frac{V_{0m}V_{m0}}{D_m}, \qquad E^{(3)} = \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{D_m D_n} - V_{00}\sum_{m\neq 0} \frac{V_{0m}V_{m0}}{D_m^2}.$$

The first two orders use the BW terms with $E_0 \to \varepsilon_0$. The third-order subtraction appears explicitly in RS and is implicit in the energy dependence of the BW denominators, as shown next.

Exercise 1 c): comparing the first few terms

Expand a Brillouin-Wigner denominator around the unperturbed one,

$$\frac{1}{E_0-\varepsilon_m}=\frac{1}{(\varepsilon_0-\varepsilon_m)+\Delta E}=\frac{1}{\varepsilon_0-\varepsilon_m}-\frac{\Delta E}{(\varepsilon_0-\varepsilon_m)^2}+\cdots, \qquad \Delta E\simeq V_{00} \text{ to this order,}$$

and insert it into the second-order BW term:

$$\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{E_0-\varepsilon_m}\simeq\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{\varepsilon_0-\varepsilon_m}-V_{00}\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{(\varepsilon_0-\varepsilon_m)^2}=E^{(2)}_{\mathrm{RS}}+\left[\text{renormalisation term of } E^{(3)}_{\mathrm{RS}}\right].$$

- Through third order the two schemes agree term by term once the BW denominators are expanded: the RS renormalisation terms *are* the expanded energy dependence of the BW denominators.
- At a *finite* order the difference is real: RS is a true polynomial in the coupling and size extensive (this week: the renormalisation terms cancel the unlinked diagrams); truncated self-consistent BW contains partial resummations to all orders and can violate additivity.
- In a small problem BW often converges faster – for the two-level model with purely off-diagonal coupling the two-vertex BW equation $E_0-\varepsilon_1=V_{12}^2/(E_0-\varepsilon_2)$ is already the exact secular equation while RS needs all even orders – while a many-body approximation must also preserve additivity for independent fragments. The RS linked expansion ensures this order by order.

Exercise 2 a): two particles in a single j-shell, Brillouin-Wigner

Pair orbitals $|m,-m\rangle$, $m=\frac{1}{2},\dots,j$: $\Omega = j+\frac{1}{2} \geq 2$ pair states, all degenerate in H_0 ($2\varepsilon_j \equiv 0$); $\langle m,-m|v|m',-m'\rangle = -G$ for *all* m,m'. In this basis

$$H = -G\,J, \qquad J = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & & \vdots \\ 1 & \cdots & 1 \end{pmatrix}_{\Omega\times\Omega}.$$

Assume $G>0$. Reference $|\phi_0\rangle = |m_0,-m_0\rangle$, $\varepsilon_0 = 0$; $V_{00} = V_{0m} = V_{mn} = -G$; every BW denominator is $E_0 - 0 = E_0 \equiv x$. The term with k vertices has k factors $-G$ and $k-1$ free sums over the $\Omega-1$ states of Q:

$$x = -G\sum_{k=1}^{\infty}\Big[\frac{-(\Omega-1)G}{x}\Big]^{k-1} = \frac{-Gx}{x+(\Omega-1)G} \implies x\,(x+\Omega G) = 0 \implies \boxed{E_0 = -\Omega G = -(j+\tfrac{1}{2})G,}$$

and at this root $(\Omega-1)G/|x| = (\Omega-1)/\Omega < 1$: the geometric series converges. The resummed form of week 44 gives it at once: $v_0 = (V_{m0})$ is an eigenvector of $J_{\Omega-1}$ with eigenvalue $\Omega-1$, so $E_0 = -G + (\Omega-1)G^2/(E_0+(\Omega-1)G)$, i.e. $E_0(E_0+\Omega G) = 0$.

A trap to avoid

For $\Omega > 2$, the fixed-point map $f(x) = -G + (\Omega-1)G^2/[x+(\Omega-1)G]$ has $|f'(-\Omega G)| = \Omega - 1 > 1$ and $|f'(0)| = 1/(\Omega-1) < 1$. The root 0 is a physical excited-state energy, outside the convergence domain of the displayed geometric series. For $\Omega = 2$, both slopes have modulus one and $f(f(x)) = x$ wherever defined. Select the negative ground-state root explicitly.

Exercise 2 b): direct diagonalisation

$J = \Omega\,|s\rangle\langle s|$ with $|s\rangle = \Omega^{-1/2}(1,\ldots,1)$ has rank one: eigenvalues Ω (once) and 0 ($\Omega - 1$ times). Hence

$$E_0 = -\Omega G = -(j+\tfrac{1}{2})G \text{ (once)}, \qquad E = 0 \ (\Omega - 1 \text{ times}), \qquad |\psi_0\rangle = \frac{1}{\sqrt{\Omega}} \sum_{m>0} |m, -m\rangle .$$

- The Brillouin-Wigner result of a) is the exact ground-state energy; the other BW root, 0, is also an eigenvalue.
- The ground state is the coherent superposition of all pair states – the *collective pair* of the seniority model. It gains the full $-\Omega G$ because the constant attractive interaction is maximally coherent in this combination; the $\Omega - 1$ orthogonal combinations (seniority two) feel no interaction at all.
- For readers who know the quasispin formalism: the seniority-zero energy for N particles in a degenerate shell is $-\frac{G}{4}N(2\Omega - N + 2)$, which at $N = 2$ gives $-G\Omega$.

Measured

$j = \frac{5}{2}, \frac{7}{2}, \frac{9}{2}$ ($\Omega = 3, 4, 5$), $G = 1$: eigenvalues $-3, -4, -5$; the two BW roots by bisection $-\Omega$ and 0 to twelve digits; plain fixed-point iteration from $-\Omega - 0.37$ ends at 0 every time (`week45.ipynb`; also `Exercises/AnswersWeek44.tex`).

Exercise 2 b): why Rayleigh-Schrödinger cannot even start, and the remedy

Rayleigh-Schrödinger

$$E^{(2)} = \sum_{m\neq 0} \frac{V_{0m}\,V_{m0}}{\varepsilon_0 - \varepsilon_m} = \sum_{m\neq 0} \frac{G^2}{0} : \text{ undefined.}$$

The reduced resolvent $Q/(\varepsilon_0 - H_0)$ does not exist: the reference is not isolated in the spectrum of H_0, which is exactly the assumption exercise 1 was built on (week 44, lecture 1, step 4). Brillouin-Wigner survives because its denominators hold the *exact* $E_0 - \varepsilon_m = -\Omega G \neq 0$: the interaction itself lifts the degeneracy, and the self-consistent equation knows it.

The remedy: degenerate perturbation theory

Enlarge the model space to the whole multiplet, $P = \sum_m |m,-m\rangle\langle m,-m|$, and diagonalise V inside it before treating anything perturbatively. Here the multiplet is the entire space, so that diagonalisation *is* the exact solution and nothing is left to expand. With several non-degenerate shells the same idea gives the multi-reference (model-space) theories and the effective interaction $H_{\text{eff}} = PHP + PHQ(E - QHQ)^{-1}QHP$ of week 44 in a model space of several determinants – the $\hat{Q}$-box of lecture 1, step 4.

Exercises, week 45

Exercises week 45 (1): the expansion, and the Hamiltonian in normal-ordered form

Let $\hat{H} = \hat{H}_0 + \hat{H}_I$, $|\Phi_n\rangle$ the eigenstates of $\hat{H}_0$ and $|\Psi_n\rangle$ those of $\hat{H}$; ground states non-degenerate, $E_0 - \varepsilon_0 = \langle\Phi_0|\hat{H}_I|\Psi_0\rangle/\langle\Phi_0|\Psi_0\rangle$, $\hat{P} = |\Phi_0\rangle\langle\Phi_0|$, $\hat{Q} = 1 - \hat{P}$.

a) Show that for any ω

$$E_0 = \varepsilon_0 + \sum_{n=0}^{\infty}\langle\Phi_0|\hat{H}_I\Big(\frac{\hat{Q}}{\omega - \hat{H}_0}(\omega - E_0 + \hat{H}_I)\Big)^n|\Phi_0\rangle.$$

b) Discuss the result for $\omega = E_0$ (Brillouin-Wigner) and $\omega = \varepsilon_0$ (Rayleigh-Schrödinger); compare the first few terms and discuss the differences.

c) Show that $\hat{H}_0 = \sum_{pq}\langle p|\hat{h}_0|q\rangle a_p^\dagger a_q = \sum_{pq}\langle p|\hat{h}_0|q\rangle\{a_p^\dagger a_q\} + \sum_i\langle i|\hat{h}_0|i\rangle$, and that

$$\hat{H}_I = \frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle a_p^\dagger a_q^\dagger a_s a_r = \frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle\{a_p^\dagger a_q^\dagger a_s a_r\} + \sum_{pqi}\langle pi|\hat{v}|qi\rangle\{a_p^\dagger a_q\} + \frac{1}{2}\sum_{ij}\langle ij|\hat{v}|ij\rangle.$$

Explain the meaning of the various symbols. Which reference vacuum has been used? Write down the diagrammatic representation of all these terms.

d) Use the diagrammatic representation of the Hamiltonian to set up all diagrams (antisymmetrised Goldstone or Hugenholtz) to second order, including the reference energy, that contribute to E_0 in Rayleigh-Schrödinger theory. Use the diagram rules to write down their closed-form expressions.

(`Exercises/ExercisesWeek45.tex`; a) and b) revisit exercise 1 of week 44 – compare your answer with the one given above.)

Exercises week 45 (2): a one-particle system drawn

Consider a one-particle system with $\hat{H} = \hat{H}_0 + \hat{H}_I$,

$$\hat{H}_0 = \sum_p \varepsilon_p a_p^\dagger a_p, \qquad \hat{H}_I = g \sum_{pq} a_p^\dagger a_q, \qquad \langle p|\hat{h}_0|q\rangle = \delta_{pq}\varepsilon_p,$$

g real, two one-particle states with $\varepsilon_1 < \varepsilon_2$; $p = 1$ is the model space, $p = 2$ the excluded space. Hole labels $ijk\ldots$ (below the Fermi level), particle labels $abc\ldots$ (above).

e) Use c) to write the Hamiltonian in normal-ordered form and set up all diagrams. Use an X for the interaction $\hat{H}_I$.

f) With the ground state (model space) $|\Phi_0\rangle = a_i^\dagger|0\rangle = a_1^\dagger|0\rangle$ and the excited state $|\Phi_i^a\rangle = a_a^\dagger a_i|\Phi_0\rangle$, $a = 2$, $i = 1$: set up the 2×2 Hamiltonian matrix, find the exact energy and expand the ground-state result in powers of g.

g) Find the ground-state energy to third order in Rayleigh-Schrödinger perturbation theory and compare with the expansion of the exact energy. Write down all diagrams which contribute and comment on your results.

Many-body connection

This is the two-state model of week 44 ($E_- = \varepsilon_1 + g - g^2/\Delta + 0\cdot g^3 + g^4/\Delta^3 + \cdots$); the new part is to *draw* it: a cross X carrying g on a line, a reference-energy bubble, and the third-order cancellation between the principal term and the renormalisation term seen as the scalar part of the middle vertex (Thursday, step zero). Answers on the week-46 slides.